

MAZE REPORT

Financial Intelligence Tracker

A Data Science Framework for Behavioural Credit Risk Modelling, Financial Stress Classification, and Savings Goal Forecasting.

Savings Goal Forecasting

Financial Stress Classification

Credit Risk Modelling

Project Name

MAZE

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Date

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Project Overview

| Problem Statement

Financial decision-making is often reactive rather than proactive due to the absence of integrated analytics systems. Institutions struggle to accurately assess risk, while individuals lack visibility into their financial health and future outcomes.

| Problem Solution

This project delivers a unified analytics platform that transforms raw financial data into actionable insights by combining stress scoring, risk prediction, and savings forecasting within a single decision-support dashboard.

Goals

- Develop a Financial Stress Scoring model to quantify individual financial pressure using behavioral indicators
- Build a Credit Risk Classification model to predict the likelihood of default
- Design a Savings Forecasting model to estimate future savings potential
- Provide actionable insights to support better financial decision-making

Core Predictive Outcomes

Financial Stress Prediction

Heuristic rule-based scoring using debt-to-income, budget usage, and emergency fund coverage.

Savings Goal Forecasting

Predicting the likelihood of reaching financial targets based on actual monthly savings behaviour.

Credit Risk Modelling

Classification of risk levels (Good, Average, Poor) derived from spending and investment habits.

The Problem with Traditional Scores

Standard credit scores often exclude individuals with limited financial history or inconsistent reporting.

MAZE shifts the paradigm from static scores to dynamic behaviour, analysing how individuals manage liquidity, expenses, and stress in real-time.

- Captures unbanked behaviour
- Real-world financial scenarios
- Reduces systemic data bias



Methodology

Building a robust automated pipeline that identifies and eliminates data leakage while prioritizing behavioural relevance.

Data Integrity

While working on the project's data visualization, I identified an inconsistency in the dataset. The financial stress level for a user is influenced by key variables such as monthly income and regular expenses. However, a problem arose where the financial stress level was recorded as low despite monthly expenses exceeding monthly income. As a result, the visualization failed to convey an accurate narrative.

Please refer to the visualization comparing the before and after scenarios below:

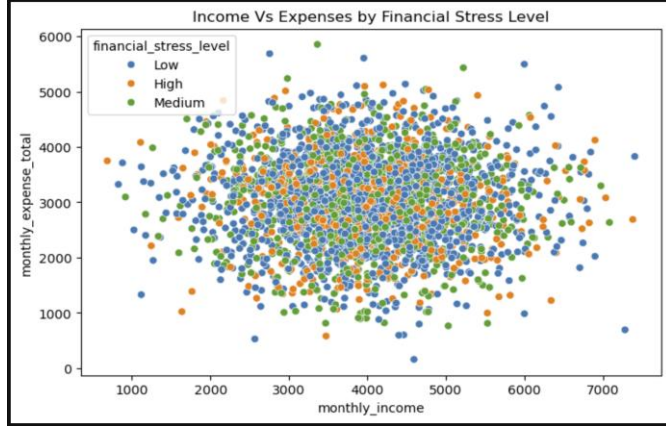


Fig A

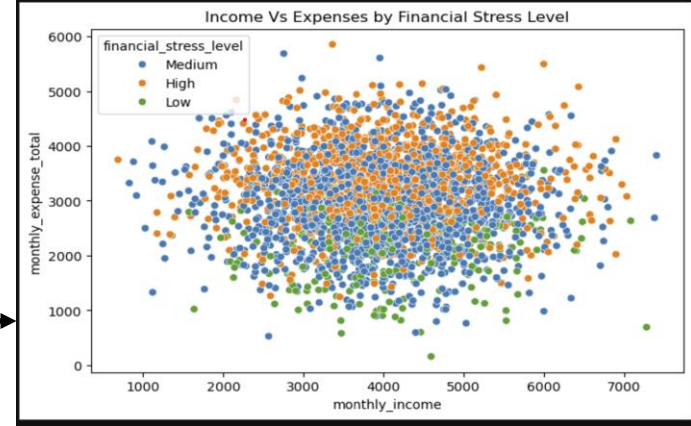


Fig B

Engineering Financial Stress

- The problem originated within the dataset. Logically, if a person spends more than their monthly income without saving, their financial stress should be categorized as high or medium; yet, the data recorded it as low, as demonstrated in the second row. To resolve this, a heuristic rule-based scoring system must be established to infer stress levels from the features, facilitating the generation of a multiclass label that distinguishes low, medium, and high stress.
- This assessment analyses individuals based on four financial metrics: debt-to-income ratio, savings rate, budget adherence, and emergency fund adequacy. Each factor contributes weighted points toward an overall stress score, with higher risk behaviours such as excessive debt, minimal savings, overspending, and insufficient emergency funds increasing the score. To avoid computational errors like division by zero, small constants are introduced in the calculations. The function is applied to each row of the dataset to create the new feature. This engineered label consolidates multiple financial indicators into a single, comprehensible category, making it valuable for analysis and as a target variable in machine learning applications.

Key Observation

The plot tells a story. The datapoints with low financial stress level(green) are settled down, the high risk are clustered up and the medium are mostly seen in the middle. There is another story to tell here, the medium points seem to be much more in number than any of the other classes which may signify class imbalance, which could be a problem when modelling

Credit Risk Insights

- Initially, bins were utilized, employing credit scores to create a multiclass system similar to the financial stress level in the dataset. However, the credit score feature in the current file does not align with the financial logic present in the other columns. If the same code were applied to genuine banking data, the model's performance would likely improve.
- In a real-world credit risk scenario, risk is assessed based on factors like elevated debt or inadequate savings. By defining "Risk" using the actual financial data within your file, the model's precision and performance will be greatly enhanced.



Credit Risk Insights Contd

Credit Score vs Loan Payment by Risk Level



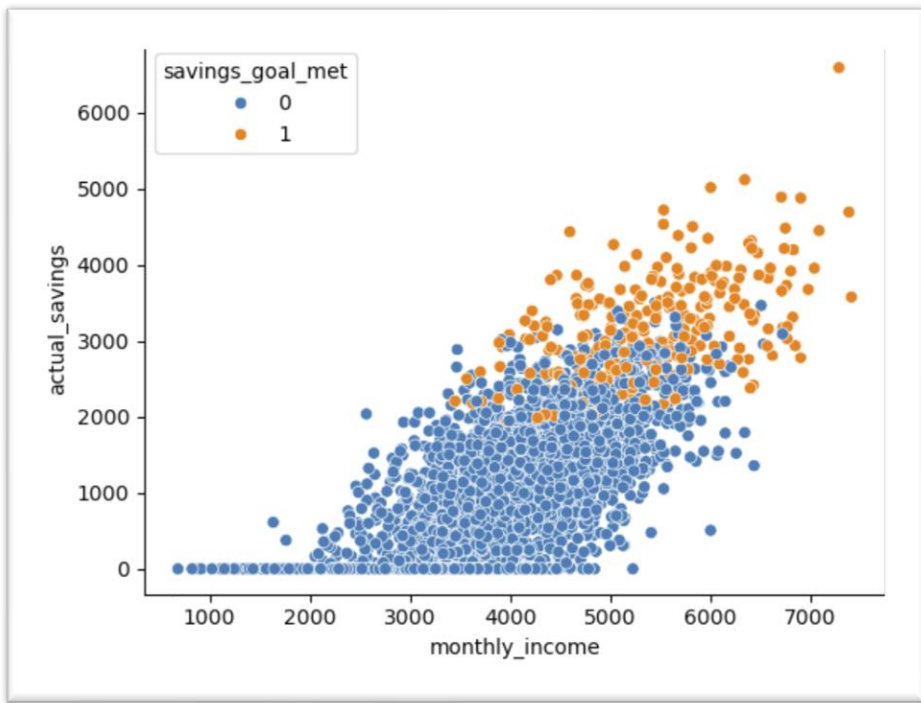
- Although measures have been taken to prevent data leakage and incorporate behavioral features, the model's performance could be enhanced by tackling class imbalance using techniques such as SMOTE to more effectively identify the minority class (class 0).
- Using advanced financial feature engineering can deepen the dataset and likely improve prediction accuracy without relying on sensitive variables like credit score and debt-to-income ratio, leading to a more robust and fair credit risk model.

Savings Goal Insights

- The dataset required minimal to no rule-based scoring for savings goal forecasting, as if it was specifically designed for that analysis. However, feature engineering was necessary to gain a more accurate understanding of the data.
- To avoid data leakage, certain critical indicators like actual savings and budget goals were excluded prior to model training.



Savings Goal Insights Contd



- **MAZE** aims not only to assess an individual's financial stress level but also to determine whether they have achieved their savings target.
- The illustration depicts the relationship between monthly income and actual savings, highlighting data points based on whether the savings goal was met.

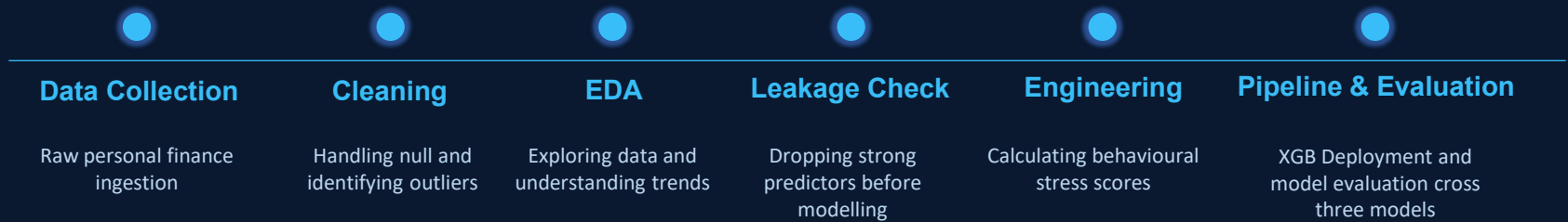
Model Accuracy Benchmarks



While Random Forest showed high accuracy, XGBoost was chosen for its balanced precision across all risk classes and robustness against class imbalance.%

Solving Data Leakage

- The initial models achieved 99% accuracy, raising concerns about potential data leakage. It was found that "Credit Score" and "Debt-to-Income Ratio" effectively served as proxies for the target variable.
- Removing these elements allows the model to focus exclusively on financial behavior, enabling its use for individuals lacking conventional banking records.



Business Impact & Deployment

Deployment Architecture

- Streamlit Frontend for user interaction.
- Joblib-serialised ML pipelines.
- Scalable cloud-ready data ingestion.

Financial Outcomes

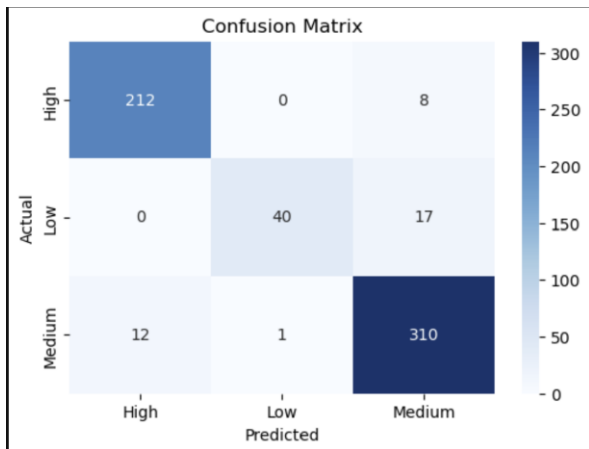
- Reduction in loan default through behavioural risk profiling.
- Increased user engagement through automated goal tracking.
- Personalized stress reduction interventions.

Business Impact Contd

- Allows financial institutions to detect high-risk clients at an early stage
- Facilitates data-informed lending choices and effective risk management
- Assists individuals in comprehending and enhancing their financial well-being
- Lays the groundwork for tailored financial advisory services

Predictive Analysis

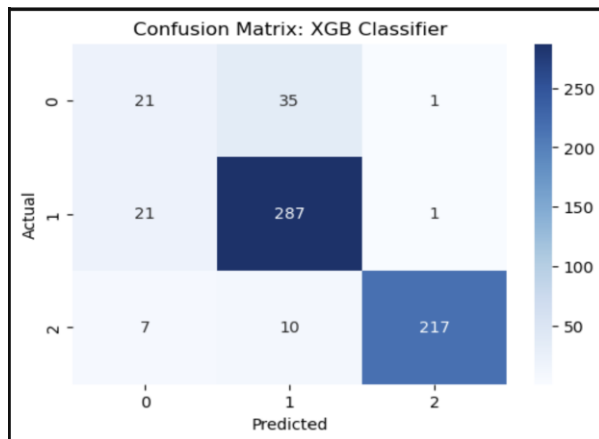
Model classification, assessment, and evaluation utilizing classification techniques and the confusion matrix.



Financial Stress Scoring (rule-based)

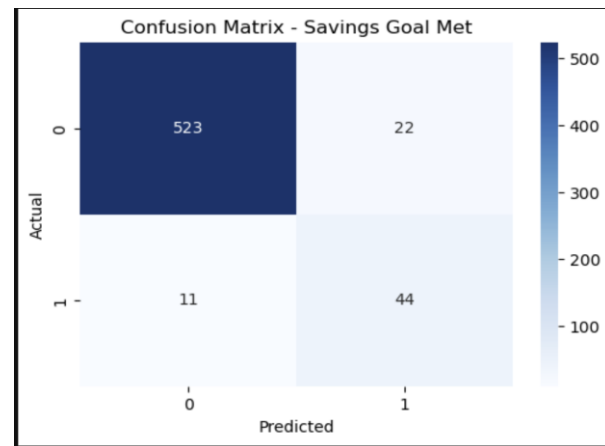
The confusion matrix shows the model's performance, with Low and High labels doing well, but the medium label causing confusion. Medium instances are often misclassified as High (False Positives).

Using an XGB Classifier may better handle these complexities and data imbalances.



Credit Risk Modelling

Initial models showed near-perfect accuracy due to data leakage from features like credit score. Removing these, the models focused on realistic behaviours like savings and investments. After retraining, XGBoost performed best with 88% accuracy and was deployed as an automated, unified pipeline.



Savings Goal Forecasting

Random Forest, Logistic Regression, and XGBoost showed similar accuracy (93–94%) but varied on minority class performance. Random Forest had high precision (0.93) and low recall (0.25); Logistic Regression had high recall (0.98) and low precision (0.57); XGBoost balanced recall (0.80) and precision (0.67), making it the best choice for deployment.

Real World Scenario

Areas for Improvement

While the model initially demonstrated strong performance, achieving up to 97% accuracy and F1-score, this success was largely driven by highly correlated financial stress indicators such as the debt-to-income ratio, along with engineered features like expense–income difference, savings delay ratio, and debt duress. To obtain a more reliable assessment of the model's capabilities, these features particularly the debt-to-income ratio, which is the most impactful predictor will be excluded in subsequent analyses. The evaluation reveals that, upon removal of these dominant features, the model's accuracy decreased substantially to 57%, providing a more realistic representation of its predictive power.

This contrast underscores the significance of prudent feature selection, since variables with excessive influence may cause inflated performance metrics and hinder generalizability. Ultimately, the results stress that although achieving high accuracy is important, developing a model that generalizes effectively and depends on balanced, independent features is essential for dependable real-world use.

Conclusion

This project demonstrates how data science can be leveraged to transform financial data into meaningful insights. By combining stress analysis, risk prediction, and savings forecasting, the system provides a comprehensive view of financial well-being.

The integration of machine learning with interactive dashboards highlights the potential for building intelligent, scalable solutions in the fintech space.

MAZE

Thank You!

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